Chapter 2
Determinants

2.2 Properties of Determinants

Determinants of Triangular Matrices

Theorem 1 If A is an $n \times n$ triangular matrix, then the determinant of A equals the product of the diagonal elements of A.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ * & a_{22} & 0 & 0 \\ * & * & a_{33} & 0 \\ * & * & * & a_{44} \end{vmatrix}$$

- By induction:
- Check that the statement is true for 1×1 or 2×2 matrices;
- Suppose the statement holds for $(n-1)\times(n-1)$ matrices (that is, the statement is true for its minors), then apply the cofactor expansion to show that the statement is still true for $n\times n$ matrices.

Determinants of Transpose

Theorem 2 If A is an $n \times n$ matrix, then $\det(A) = \det(A^T)$.

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{n1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{n2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{vmatrix}$$

- By induction:
- Check that the statement is true for 1×1 or 2×2 matrices;
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Properties of Determinants

Theorem 3

- If A has a row or column consisting entirely zeros, then $\det(A) = 0$.
- If A has two identical rows or columns, then $\det(A) = 0$.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{vmatrix}$$

Properties of Determinants

Lemma 1

Lemma: Let A be an $n \times n$ matrix. If A_{jk} denotes the cofactor of a_{jk} for $k = 1, \ldots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A) & if \quad i = j \\ 0 & if \quad i \neq j \end{cases}$$

Proof

- If $i = j$, then the left side is just the cofactor expansion of $\det(A)$ along the ith row.
- If $i \neq j$, let A^* be the matrix obtained by replacing the jth row of A by the ith row of A:

$$A^* = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{i1} & \cdots & a_{in} \\ \cdots & \cdots & \cdots \\ a_{i1} & \cdots & a_{in} \\ \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{matrix} \\ \\ j^{th} \text{ row} \\ \\ i^{th} \text{ row} \\ \\ \\ \end{matrix}$$

- By Theorem 3, $\det(A^*) = 0$
- Expand $\det(A^*)$ along j^{th} row:

$$\det(A^*) = a_{i1}A^*_{j1} + a_{i2}A^*_{j2} + \cdots + a_{in}A^*_{jn}$$
$$= a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn}$$

Properties of Determinants

Row Operation III : *A multiple of one row is added to another row.*

- Let E be the elementary matrix of type III formed from I by adding c times the ith row to the jth row, then

$$\det(EA) = \det(A)\,.$$

- In particular, E is a triangular matrix and all diagonal elements are 1, therefore $\det(E) = 1$, and

$$\det(EA) = \det(A) = \det(E)\det(A).$$

Proof

- If $\det(EA)$ is expanded by cofactors along j^{th} row:

$$\begin{aligned}\det(EA) &= (ca_{i1} + a_{j1})A_{j1} + \cdots + (ca_{in} + a_{jn})A_{jn} \\ &= (a_{j1}A_{j1} + \cdots + a_{jn}A_{jn}) + c(a_{i1}A_{j1} + \cdots + a_{in}A_{jn}) \quad \textit{(Lemma 1)} \\ &= (a_{j1}A_{j1} + \cdots + a_{jn}A_{jn}) = \det(A)\end{aligned}$$

Properties of Determinants

Row Operation I: *Two rows of A are interchanged*

- If A is an $n \times n$ matrix and E_{ij} is the $n \times n$ elementary matrix formed by interchanging the ith row and j^{th} row of I, then
$$\det(EA) = -\det(A).$$
- In particular, $\det(E_{ij}) = \det(E_{ij}I) = -\det(I) = -1$, then
$$\det(EA) = -\det(A) = \det(E)\det(A).$$

Proof (hint)

- By induction
- Expand the determinant along an unchanged row.

Properties of Determinants

Row Operation II : A *row of A is multiplied by a nonzero constant.*

- Let E be the elementary matrix of type formed by multiplying the ith row by a nonzero constant α, then

$$\det(EA) = \alpha \det(A).$$

- In particular, $\det(E) = \det(EI) = \alpha\det(I) = \alpha$, then

$$\det(EA) = \alpha\det(A) = \det(E)\det(A).$$

Determinant and elementary matrices

In summary, if E is an elementary matrix, then

$$\det(EA) = \det(E)\det(A)$$

where

$$\det(E) = \begin{cases} -1 & \text{if } E \text{ is of type I} \\ \alpha \neq 0 & \text{if } E \text{ is of type II} \\ 1 & \text{if } E \text{ is of type III} \end{cases}$$

Similar results hold for column operations:

$$\det(AE) = \det(A)\det(E)$$

because if E is an elementary, then E^T is also an elementary matrix and

$$\det(AE) = \det\left((AE)^T\right) = \det\left(E^T A^T\right) = \det\left(E^T\right)\det\left(A^T\right)$$
$$= \det(A)\det(E)$$

Determinant and inevitability

Theorem

An $n \times n$ matrix A is singular if and only if $\det(A) = 0$.

Proof

- Matrix A can be reduced to row echelon form with a finite number of row operations. Thus,

$$U = E_k E_{k-1} \dots E_1 \mathrm{A}$$

- Then,

$$\det(U) = \det(E_k)\det(E_{k-1}) \dots \det(E_1)\det(A)$$

- Therefore, $\det(A) = 0$ if and only if $\det(U) = 0$. If A is nonsingular, then U is triangular with all diagonal equals to 1, $\det(U) = 1$. If A is singular, U contains a row of all zeros, $\det(U) = 0$.

Properties of Determinants

Theorem 4

If A and B are $n \times n$ matrices, then

$\det(AB) = \det(A)\det(B)$.

Proof

- Suppose B is singular matrix, then $B\boldsymbol{x} = 0$, has nontrivial solutions, then $\boldsymbol{x}$ is also a nontrivial solution to $AB\boldsymbol{x} = 0$. Then AB is also a singular matrix. So $\det(AB) = 0 = \det(A) \cdot \det(B) = \det(A) \cdot 0 = 0$.
- If B is not singular, then is row equivalent to the identity matrices or $B = E_1E_2 \cdots E_k$ for some k. And $\det B = \det(\mathrm{E}_1)\det(E_2) \cdot \det(E_k)$. (This is due to the property of elementary matrices)
- $\det(AB) = \det(AE_1E_2 \cdots E_k) = \det(A)\det(E_1) \cdot \ldots \cdot \det(E_k) = \det(A) \cdot \det(B)$

Exercise

For a square matrix A, what can you conclude about $\det(AA^T)$?

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Solution

$$\det(AA^T) = (\det(A))^2$$

Properties of Determinants

1. If A is nonsingular, then det(A)≠0 and $\det(A^{-1}) = \frac{1}{\det(A)}$
2. $\det(kA) = k^n \det(A)$

Proof

$$|A| \cdot |A^{-1}| = |A \cdot A^{-1}| = |I| = 1 \quad \Rightarrow \quad |A^{-1}| = \frac{1}{|A|}$$

$$A \underset{kR1,\, kR2,\, \ldots,\, kRn}{\approx} kA \quad \Rightarrow \quad |kA| = k^n |A|$$

Compute determinant

- **Example 1** Compute det *A*, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

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- **Solution** The strategy is to reduce *A* to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries. Type III row operations do not change the determinant:

$$|A| = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

$R2 + 2R1$ $\qquad$ $R3 + R1$

Compute determinant

- An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$detA = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

Example

Evaluate the determinant of $\begin{vmatrix} 3 & -1 & 2 \\ 4 & -6 & 9 \\ -2 & 3 & -3 \end{vmatrix}$

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Solution

$$\begin{vmatrix} 3 & -1 & 2 \\ 4 & -6 & 9 \\ -2 & 3 & -3 \end{vmatrix} \underset{\text{R2+2R3}}{=} \begin{vmatrix} 3 & -1 & 2 \\ 0 & 0 & 3 \\ -2 & 3 & -3 \end{vmatrix} = (-3)\begin{vmatrix} 3 & -1 \\ -2 & 3 \end{vmatrix} = -21$$

A simpler way to find the determinant.

Example

- Find |A| for $A = \begin{bmatrix} -2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & -6 & -8 \end{bmatrix}$

Example

If A is a 2×2 matrix with $|A| = 4$, use Theorems to compute the following determinants.

$(a)|3A|$ $(b)|A^2|$ $(c)\left|5A^T A^{-1}\right|$, assuming A^{-1} exists

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Solution

(a) $|3A| = (3^2)|A| = 9 \times 4 = 36.$

(b) $|A^2| = |AA| = |A|\,|A| = 4 \times 4 = 16.$

(c) $|5A^T A^{-1}| = (5^2)|A^T A^{-1}| = 25|A^T||A^{-1}| = 25$

Exercise

For $C = \begin{bmatrix} k & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & k \end{bmatrix}$, find all values of $k \in R$, such that $Cx = 0$ has a nontrivial solution.

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Solution

$k = 0 \; or \; k = 1$

Exercise

For $D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, find an equation that λ satisfies so that $(D - \lambda I)x = 0$ has a nontrivial solution.

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Solution

$$\lambda^2 - (a + d)\lambda + ad - bc = 0$$

Exercise

Let x and y be vectors in R^n with $n > 1$. Prove that if $A = \mathbf{x}\mathbf{y}^T$ then $\det(A) = 0$.

Proof

$$A = \begin{bmatrix} a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{bmatrix}$$

Perform row operations of type II.

$$|A| = a_1 \begin{vmatrix} b_1 & b_2 & \dots & b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{vmatrix} = a_1a_2 \begin{vmatrix} b_1 & b_2 & \dots & b_n \\ b_1 & b_2 & \dots & b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{vmatrix} = 0$$

Because the first two rows are identical.

True or False

- $\det(AB) = \det(BA)$
- $\det(A + B) = \det(A) + \det(B)$
- $\det(cA) = c \det(A)$
- $\det((AB)^T) = \det(A) \det(B)$
- $\det(A) = \det(B) \Rightarrow A = B$
- $\det(A^k) = \det(A)^k$
- A triangular matrix is nonsingular if and only if its diagonal entries are all nonzero.